

Table Types



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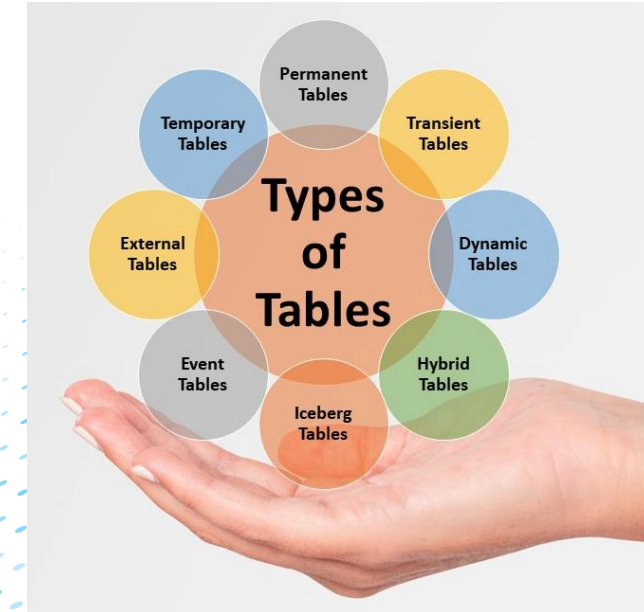
Section Introduction–Snowflake Table Types

- What are the table types supported in Snowflake ?
- LAB 1–Create tables in Snowflake.
- LAB 2–Impact of Temporary tables on a Permanent table
- Summary.

What are the Table Types supported in Snowflake?

Snowflake supports 8 different Table Types.

1. Permanent Tables.
2. Transient Tables.
3. Temporary Tables.
4. Dynamic Tables.
5. External Tables.
6. Hybrid Tables.
7. Event Tables.
8. Iceberg Tables.



What are the Table Types supported in Snowflake?

Permanent Tables

- Permanent Tables are the **default** table type in Snowflake.
- Permanent Tables are created with the **CREATE TABLE <tablename>** syntax
- Permanent Tables can have a maximum data retention period of 90 days with the Enterprise Edition and higher. For the Standard Edition this will be a maximum of 1 day.
- Permanent Tables support Fail-Safe retention period of 7 days.
- Permanent Tables are best suited for tables that need to retain data permanently like FACTS and DIMENSIONS.
- When a Database or Schema is created it is a Permanent Database or Schema by default.



What are the Table Types supported in Snowflake?

Transient Tables

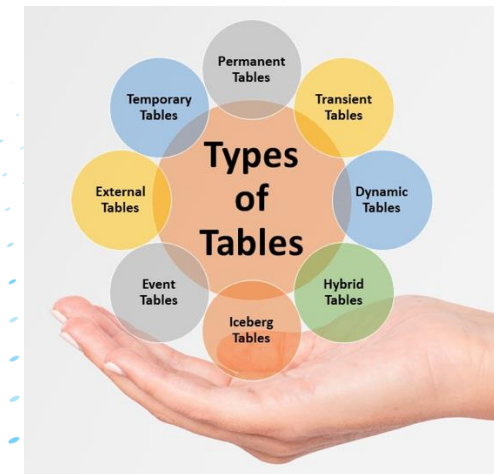
- Transient Tables are created with the keyword **TRANSIENT** in the **CREATE TRANSIENT TABLE <tablename>** syntax.
- Transient Tables can have maximum of 1 day of Time Travel irrespective of the Snowflake edition.
- Transient Tables are used in those cases where data in the tables are stored temporarily and deleted within a few days.
- Transient Tables persist until the table is dropped.
- Fail-Safe storage is not supported in Transient tables.
- Databases and Schemas can be created as Transient Database or Transient Schema, all tables created inside a Transient Database or Schema is Transient by default.



What are the Table Types supported in Snowflake?

Temporary Tables

- Temporary Tables are created with the keyword **TEMPERORY** in the **CREATE **TEMPERORY** TABLE <tablename>** syntax.
- Temporary Tables exist only during the session they are created in. Once the session ends the Temporary Tables are dropped and all data is permanently deleted.
- Temporary Tables are best suited to temporally store data that is required within a session like a Stored Procedure execution to store data for intermediate calculation.
- Temporary Tables support Time Travel and can have a maximum data retention period of 1 day.
- Fail-Safe storage is not supported in Temporary tables.
- Temporary Databases and Schema cannot be created in Snowflake.



What are the Table Types supported in Snowflake?

Permanent	Transient	Temporary
Upto 90 day Time Travel	Upto 1 day Time Travel	Upto 1 day Time Travel
Fail-Safe	No Fail-Safe	No Fail-Safe
Create Table <TableName>	Create TRANSIENT Table <TableName>	Create TEMPORARY Table <TableName>
Works best for tables that need to store data permanently E.g: Facts and Dimension.	Works best for tables that need to store data around a day E.g: Staging tables in DWH.	Temporary tables are best suited to temporarily store data that is required within a session like a stored procedure which is calculating something and needs to store intermediate results in a place for a short duration of time.
Create Database <databasename>	Create TRANSIENT Database <databasename>	TEMPORARY Database cannot be created.
Create SCHEMA <databasename>	Create TRANSIENT SCHEMA <databasename>	TEMPORARY SCHEMA cannot be created.
Create TABLE <tablename>	Create TRANSIENT TABLE <tablename>	Create TEMPORARY TABLE <tablename>

Summary

- Snowflake supports 8 different Table Types
 - Permanent Tables.
 - Transient Tables.
 - Temporary Tables.
 - Dynamic Tables.
 - External Tables.
 - Hybrid Tables.
 - Event Tables.
 - Iceberg Tables.
- Permanent Tables are the **default** table type in Snowflake.
- Permanent Tables are created with the **CREATE TABLE <tablename>** syntax.
- Permanent Tables can have a maximum data retention period of 90 days with the Enterprise Edition and higher. For the Standard Edition this will be a maximum of 1 day.
- Permanent Tables support Fail-Safe retention period of 7 days.
- Permanent Tables are best suited for tables that need to retain data permanently like FACTS and DIMENSIONS.
- Permanent Tables must be used if you need Time Travel data for recovery greater than 1 day.
- When a Database or Schema is created it is a Permanent Database or Schema by default.



Summary

- Transient Tables are created with the keyword **TRANSIENT** in the **CREATE TRANSIENT TABLE <tablename>** syntax.
- Transient Tables can have maximum of 1 day of Time Travel irrespective of the Snowflake edition.
- Transient Tables are used in those cases where data in the tables are stored temporarily and deleted within a few days.
- Fail-Safe storage is not supported in Transient tables.
- Databases and Schemas can be created as Transient Database or Transient Schema, all tables created inside a Transient Database or Schema is Transient by default.
- Transient Tables can have maximum of 1 day of Time Travel irrespective of the Snowflake edition.
- Transient Tables are used in those cases where data in the tables are stored temporarily and deleted within a few days.
- Transient Tables should be used in all those cases where data can be easily recreated from the source.
- Fail-Safe storage is not supported in Transient tables.



Summary

- Temporary Tables are created with the keyword **TEMPERORY** in the **CREATE **TEMPERORY** TABLE <tablename>** syntax.
- Temporary Tables exist only during the session they are created in. Once the session ends the Temporary Tables are dropped and all data is permanently deleted.
- Temporary Tables are best suited to Temporarily store data that is required within a session like a Stored Procedure execution or SQL script to store data for intermediate calculation.
- Temporary Tables support Time Travel and can have a maximum data retention period of 1 day.
- Fail-Safe storage is not supported in Temporary tables.
- Temporary Databases and Schema cannot be created in Snowflake.
- A Temporary Table and permanent table can share the same name and be created inside the same schema.
- Temporary Table will take precedence inside the session if it shares the same name with a permanent table.
- Any DML or DDL statement executed in the schema will be directed to the Temporary Table.
- There is no way to bypass the Temporary Table and access the Permanent Table except drop it.



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